■ 1.8.6 Advanced Topic: Parametric Plots

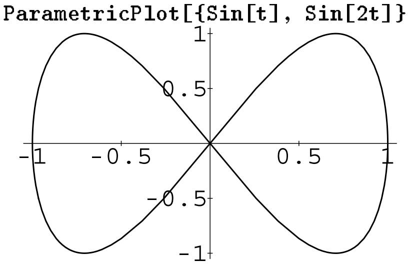
Section 1.8.1 described how to plot curves in *Mathematica* in which you give the y coordinate of each point as a function of the x coordinate. You can also use *Mathematica* to make *parametric* plots. In a parametric plot, you give both the x and y coordinates of each point as a function of a third parameter, say t.

`ParametricPlot[{`f_x`, `f_y`}, {`t`, `$tmin$`, `$tmax$`}]`	make a parametric plot
`ParametricPlot[{{`f_x`, `f_y`}, {`g_x`, `g_y`}, ...}, {`t`, `$tmin$`, `$tmax$`}]`	plot several parametric curves together
`ParametricPlot[{`f_x`, `f_y`}, {`t`, `$tmin$`, `$tmax$`}, AspectRatio -> Automatic]`	attempt to preserve the shapes of curves

Functions for generating parametric plots.

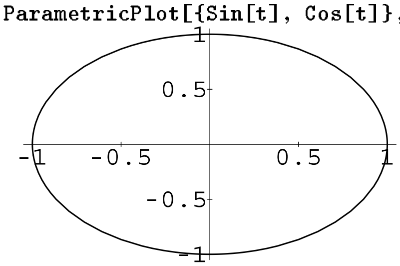
Here is the curve made by taking the x coordinate of each point to be `Sin[t]` and the y coordinate to be `Sin[2t]`.

In[1]:= `ParametricPlot[{Sin[t], Sin[2t]}, {t, 0, 2Pi}]`



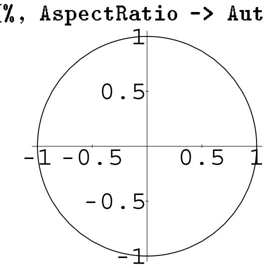
The "shape" of the curve produced depends on the ratio of height to width for the whole plot.

In[2]:= `ParametricPlot[{Sin[t], Cos[t]}, {t, 0, 2Pi}]`



Setting the option `AspectRatio` to `Automatic` makes *Mathematica* preserve the "true shape" of the curve, as defined by the actual coordinate values it involves.

```
In[3]:= Show[%, AspectRatio -> Automatic]
```



`{`r`[`t`] Cos[`t`], `r`[`t`] Sin[`t`]}`	polar plot with radius r`[`t`]` at angle t
`{Re[`f`], Im[`f`]}`	phase or Argand diagram of a complex function
`{Log[`f`], Log[`x`]}`	log-log plot

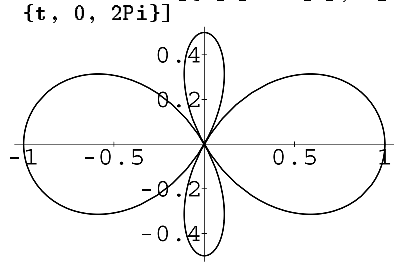
Some types of parametric plots.

This defines a function `r`.

```
In[4]:= r[t_] := (3 Cos[t]^2 - 1)/2
```

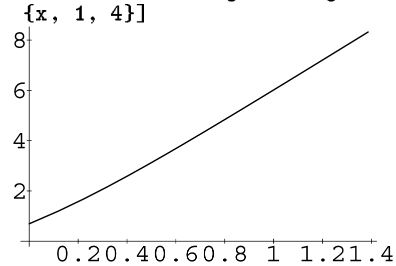
This gives a plot in polar coordinates, with `r[t]` as the radius function at angle `t`.

```
In[5]:= ParametricPlot[{r[t] Cos[t], r[t] Sin[t]},
            {t, 0, 2Pi}]
```



This makes a log-log plot of $x^2 + x^6$ as a function of x.

```
In[6]:= ParametricPlot[{Log[x], Log[x^2 + x^6]},
            {x, 1, 4}]
```



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